



Assessment and prediction of environmental sustainability: novel grey models comparative analysis of China vs. the USA

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Abstract

Emission forecasting is vital for policy-making and emission reduction goals globally. This research aimed to perform an accurate model for forecasting and assessing CO₂ emissions and the production of renewable electricity for the top two countries contributing to these emissions, the USA and China. In this study, we employed three novel advanced mathematical grey models: optimized discrete grey model (ODGM), nonhomogeneous discrete grey model (NDGM), and variable speed and adaptive structure grey model (VSSGM) to estimate the future trends of CO₂ emissions and renewable electricity production. These breakthrough models added value in this field of research by reducing uncertainty surrounding ambiguity and numerical ranges and improving accuracy in assessments by using small samples and imperfect information. The findings showed that, by 2026, China's electricity production based on renewable sources would be higher than that of the USA. We find CO₂ emissions in a downward trend, with more significant reductions in the USA than in China by the year 2026. The contributions of this study are the application of novel VSSGM and the use of synthetic relative growth rate modeling for predicting the overall growth of CO₂ emissions and the production of renewable electricity in analyzed countries. The originality of this study lies in proposing a novel synthetic doubling time model to compute how long it will take, for China and the USA, to reduce their CO₂ emissions and doubling their increase in renewable electricity production.

Keywords Environmental sustainability · Prediction · Carbon dioxide (CO₂) emissions · Renewable energy · Emission forecasting · Grey models

Highlights

- The modeling in this study improves our ability to predict energy and emissions between the two largest countries contributing to GHG emissions.
- Synthetic relative growth rate modeling predicts the overall growth of CO₂ emissions and the production of renewable electricity.
- The use of a novel synthetic doubling time model to compute how long it will take for China and the USA to reduce CO₂ emissions.
- By 2026, China's electricity production by renewable sources will be higher than that of the USA.

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Introduction

A global increase in the rate of carbon dioxide (CO₂) emissions caused by increased industrialization and heightened generation of electricity through non-renewable energy sources is, in turn, accelerating global warming (Ko et al. 2017). It appears that high CO₂ emitting countries may only be concerned with expanding their economies. At the same time, experiencing population growth and these changes comes at the expense of environmental sustainability. Actions need to be taken, primarily related to the high CO₂ emitting countries, in order to reduce waste in the form of CO₂ emissions and the subsequent effects of global warming (Paramati et al. 2017; Shahbaz et al. 2016). A major cause of CO₂ emissions is the expansion of the agriculture sector as it meets the needs of an increasing population and increasing greenhouses (GHGs) generation. These GHGs are notorious in producing detrimental impacts on the environment, and whose major component is CO₂ in sum with other gases normalized to a CO₂ equivalent (Rouholamini et al. 2020; Yildiz and Karan 2020; Ikram et al. 2020b).

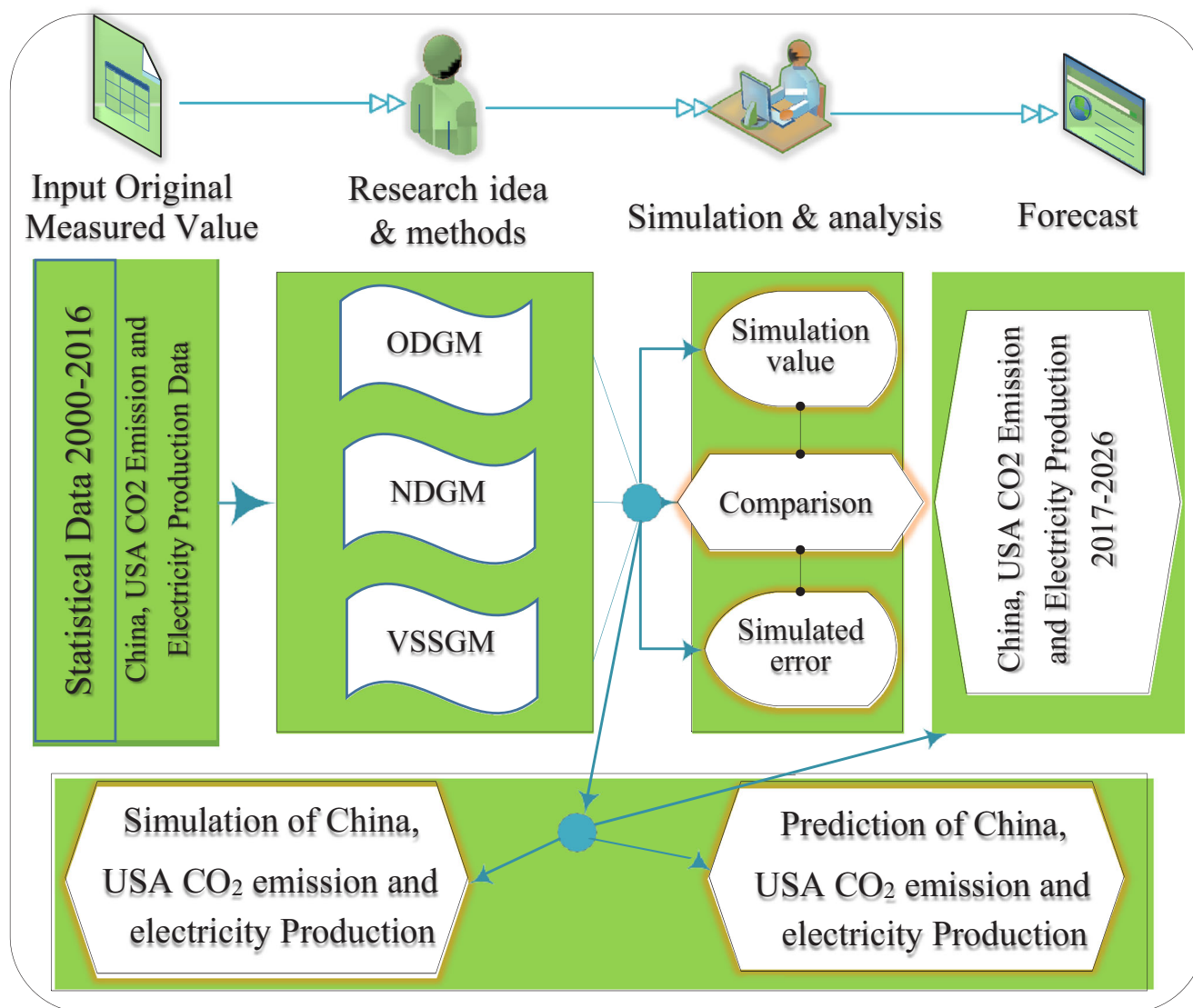


Fig. 1 Research framework based on novel grey models

As proposed by the Kyoto Protocol framework, annual reductions of these GHGs annually are necessary to try and head off a 2 °C increase in global temperature to meet the required goals to help stop climate changes (Özdingiş et al. 2018; Chen et al. 2020; Shahbaz et al. 2019a, b, c; Durán-Romero et al. 2020). Another primary source of CO₂ is deforestation and the overconsumption of fossil fuels from increased industrialization. Over the last two decades, global warming and the increased threat of climate change have reached alarming levels worldwide (Hansen and Sato 2016; Shahbaz et al. 2019a). The International Energy Outlook Report estimates a growth rate of approximately 7.6% of CO₂ to hit 43.2B metric tons by the year 2040 and up from 35.6B metric tons in the year 2020 (Zhang et al. 2015; Amin et al. 2020).

Due to an increasing amount of important country level developments, there is a critical need for more research and

modeling of emissions and energy production. This research can enable decision making for practitioners and governments and provide new research opportunities. Scholars have approached this opportunity in the past by using qualitative and quantitative methodologies for predicting emissions and renewable energy production (see, for example, Hobman and Frederiks (2014), Lee and Huh (2017), Daim et al. (2012), Kowalski et al. (2009), and Moonchai and Chutsagulprom (2020)). In this study, we conduct an empirical investigation aiming to test multiple models and future impacts. We chose this modeling approach, in part, to see if it can help confirm or reject prior arguments.

To this end, we propose and use advanced Optimized Discrete Grey Model - ODGM, Nonhomogeneous Discrete Grey Model - NDGM, and Variable Speed and Adaptive Structure Grey Model - VSSGM, to forecast renewable electricity production and CO₂ emissions (Xie et al. 2015). All three

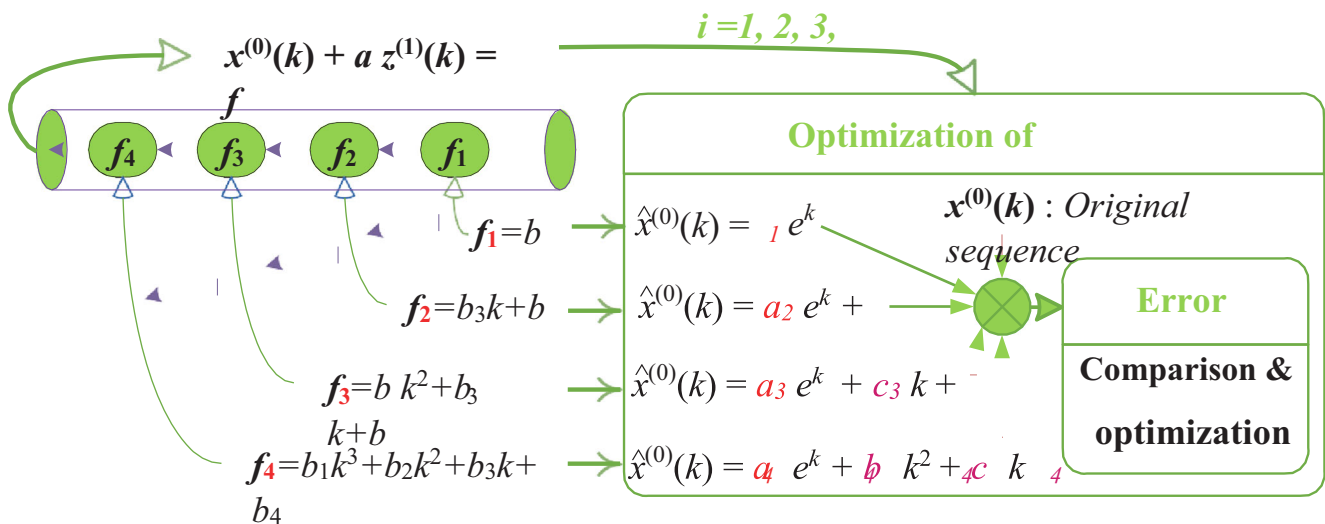


Fig. 2 VSSGM model with optimized characteristics

models provide an opportunity to increase forecasting accuracy over prior modeling. This increased accuracy comes as prior studies were not able to accurately describe results given the features of a small sample size and proposed estimation methods. These prior methods require detailed information on various socioeconomic scenarios (Knüppel 2014; Mazzeu et al. 2018). Therefore, new contributions from this study include the ability to handle increased complexity. At the same time, production activities, and the standards of living at the country level can also be integrated into our modeling (Weron 2014). With the complexity of this modeling, there are a number of important variables and prior studies we can build upon.

Due to the rapid acceleration of economic development, energy consumption is growing in fast-growing countries such as China. This growth causes higher rates of CO₂ and GHG emissions. In this scenario, various approaches can be used to predict CO₂ emissions with excellent outcomes. These approaches considered the use of artificial intelligence (AI), spatial econometrics, and series analysis, to name a few (García-Martos et al. 2013; Chen and Yang 2015; Sun et al. 2017). Large samples are used in these methods. To reduce uncertainty, the data needs to follow statistical assumptions (Feng et al. 2014). The GM (1,1) and GM (1,N) models were commonly used to estimate the CO₂ emissions since both models are characterized as “grey models.” In

the research of Li et al. (2020a, b), it is found the application of a nonlinear grey model named “Bernoulli NGBM” (1,1) can be applied for forecasting CO₂ emissions from China. These authors proposed a more accurate iterative numerical method in order to optimize the parameters used in their modeling. The research of Wu et al. (2014) forecast CO₂ emissions for BRICS countries with the application of a multivariable gray model considering the growth of gross domestic product (GDP). Their work is interesting in that it also included a forecast for energy consumption. Another forecast of CO₂ emissions in China comes from Ding et al. (2017). The authors used another grey model that was integrated with an optimization algorithm and considering data from fuel combustion, GDP, the urban population, and energy consumption. In the same vain of CO₂ emissions forecasting, Wang and Li (2019) tested a nonlinear grey model based on multiple variables based on fossil fuel-based energy consumption and GDP.

China’s National Bureau of Statistics (NBS) stated that, in 2017, total energy consumption increased 2.9% when compared with 2016. It represents an increase of 4.36 gigatons (Yang and Shi 2017). China is aiming to reduce the energy-intensive consumption (measured as energy consumption per unit of GDP) at 3.4%, with a reduction of 3.7% in 2017, and the share of coal in total energy consumption declined by 1.7 % (Enerdata 2017). While the gas, hydropower, nuclear, and wind energy increased by 1.5% in 2017, the electricity consumption increased by 6.6%. The Chinese National Energy Administration (NEA) found that in 2017, there was a 5.5% growth in industrial power consumption. The installed power generation capacity of the country grew by 137 GW to above 1800 GW, during the year, with nearly 13 GW of hydropower capacity and nearly 46 GW of thermal capacity installed (IEA 2017). In the USA, from all types of energy, about 82% came from fossil fuel sources (Meschi and Norheim-Hansen 2020). The largest of the US sources of energy in 2014 came from

Table 1 Accuracy level of MAPE %

MAPE %	Forecasting accuracy
Higher than 50 %	Poor prediction accuracy
20–50 %	Moderate
10–20 %	Good
Lower than 10%	Highly accuracy level

Source: (Lewis 1982).

petroleum (35%) and by natural gas (28%), followed by coal (18%) and nuclear power (8%). Renewable sources contributed only 10% of these energy sources (Rahman et al. 2017; Shahbaz et al. 2019c). This data from the USA and China set the stage for helping predict future energy and emissions from these global superpowers.

Our research aims to develop an accurate model for CO₂ emissions and the use of renewable electricity forecasting. Our modeling of the top two CO₂ emitters globally, the USA and China, take advantage of ODGM, NDGM, and VSSGM grey modeling. These models are useful for this kind of quantitative study as they help to minimize uncertainty, ambiguity, and data bias. The application of these models eliminates uncertainty and bias by offering more in-depth analyses of renewable energies and emissions. Outcomes of this study provide future trends in the amount of electricity generation and CO₂ emissions in order to help policy-makers in decision-making processes regarding the future expected outcomes. Outcomes of the study can also advance research with the application of competing forecasting models. Our empirical methodology utilizes available data making replication of this study and future work more accessible to both practitioners and scholars. The outcomes of this study are made more reliable by the use of information from relevant authorities, namely, weather and climate scientists, governments, and other researchers in this field.

This study is structured as follows. The “Research framework and methods” section discusses the development steps of ODGM, NDGM, and VSSGM models. The “Results and discussion” section presents the results with discussions and contributions to the field, and finally, we present conclusions in the last section. New insights and contributions of this study came from the use of a relative growth rate (RGR), doubling time (D_t) modeling, and performance criteria methods to estimate future generation of renewable electricity and CO₂ emission proposed by Javed and Liu (2018) and successfully

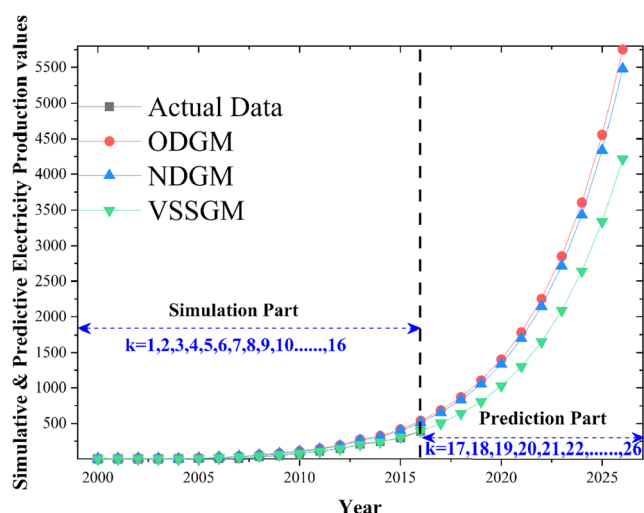


Fig. 3 China's simulation and predictive values of electricity production from 2000 to 2026

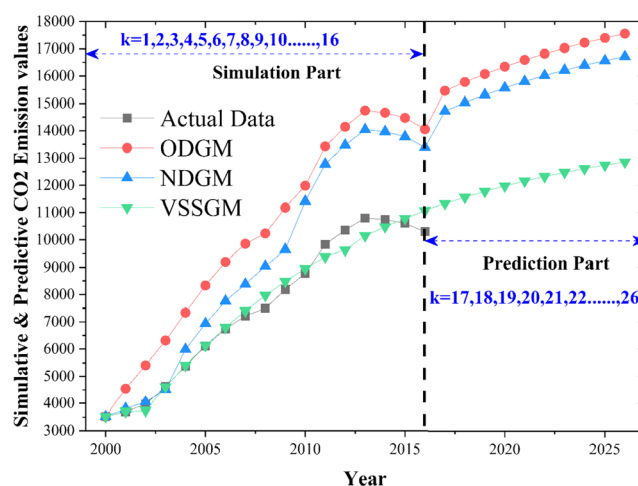


Fig. 4 China's simulation and predictive values of CO₂ emissions from 2000 to 2026

utilized by Ikram et al. (2019). The methodology suggested for this study is a breakthrough in forecasting the generation of both renewable electricity energy and emissions. This validated method is applicable due to the significant value it can add to renewable sources and CO₂ emissions forecasting studies.

Research framework and methods

The electricity and CO₂ emission data are from the official website of International Energy Statistics (EIA). For our data collection, China and the USA were looked at from 2000 through 2016, as the highest countries in CO₂ emission and electricity consumption from renewable energy sources globally. For this study, we propose mathematical modeling using a Grey forecasting model. This forecasting model has been proven effective for studies such as this, and its accuracy is enabled with the use of publicly available data (Zhao et al. 2012; Ikram et al. 2019). The ODGM, NDGM, and VSSGM are proven approaches for application to small data sets and are, therefore, useful tools for this study (Ma and Liu 2017). The framework for our research regarding China and the US renewable electricity production and CO₂ emission prediction is in Fig. 1.

Optimized discrete grey model (ODGM)

The following five steps operationalize the ODGM forecasting model. We discuss these steps in sufficient detail to help other researchers replicate this approach.

Step 1: The model testing starts with the description of ODGM, based on mathematical modeling adopted to optimize the discrete grey model. This optimization is comprised of objective functions. The advantages of grey forecasting models are to show high accuracy for uncertain data systems and incomplete

Table 2 Homogeneous exponential sequence X1 parameters

Model	Exponential sequence	Parameters	Models structuring
NDGM	Homogeneous	$\hat{\beta}_1 = (1.7, 1.63)^T$	$\hat{z}^{(1)}(k+1) = 1.7z^{(1)}(k) + 1.63$
ODGM	Homogeneous	$\hat{\beta}_2 = (1.7, 0.0, 1.63, 0.0)^T$	$\hat{z}^{(1)}(k+1) = 1.7z^{(1)}(k) + 0.0k + 1.63, \hat{z}^{(1)}(1) = z^{(1)} + 0.0$
VSSGM	Homogeneous	$a_3 = (-0.61143, 0.0, 0.0, 0.0, 1.1542)^T$	$\hat{z}^{(0)}(k) - 0.61143z^{(1)}(k) = 0.0k^{(3)} + 0.0k^{(2)} + 0.0k + 1.1542$

information for the design of the variables in the forecast (Ma and Bin 2016). Next, CO₂ emission and renewable electricity production data were used from 2000 to 2016 with the total amount of emissions as an original sequence during the development of the ODGM model (see Eq. 1).

$$Z^{(0)} = \{z^{(0)}(1), z^{(0)}(2), \dots, z^{(0)}(m)\} \quad (1)$$

Step 2: The minimum orthogonal array level is developed in the second step of ODGM model construction. This is done while taking into account the inner array variables. Further, a repetitive trail approach is needed for the outer array of variables to make and refine the essence performance of function through fluctuations. An acquired sequence of operations can then be drawn, as seen in Eq. 2.

$$Z^{(1)} = \{z^{(1)}(1), z^{(2)}(2), \dots, z^{(1)}(n)\} \quad (2)$$

Taking this a step further involves the transformation of carbon emission and renewable electricity production sequences presented by $Z^{(1)}$, and this can be written as follows.

$$Z^{(1)}(l) = \sum_{i=1}^l Z^{(0)}(i), \quad l = 1, 2, \dots, m \quad (3)$$

Step 3: With the help of Eq. 3, the coefficient of relation can be calculated to develop ODGM based on accumulated weights of relation coefficient and value of the prediction sequence. Prior studies confirmed that, at the construction of the prediction model, the coefficient weight is derived from principal component

analysis of variable function. Carbon emission and electricity generation by renewable sources are a parameter series such that $\hat{\beta} = (\hat{\beta}_1, \hat{\beta}_2)^T$ and the following equation can be written as follows.

$$B = \begin{bmatrix} z^{(1)}(1) & 1 \\ z^{(1)}(2) & 1 \\ \vdots & \vdots \\ z^{(1)}(m-1) & 1 \end{bmatrix}$$

$$Z^{(1)}(l+1) = \beta_1 z^{(1)}(l) + \beta_2, \quad l = 1, 2, \dots, m-1 \quad (4)$$

A least square method is applied to compute the ODGM series parameters in Eq. 5, and further, the formula of $\hat{\beta}$ is presented in Eq. 6.

$$Z^{(1)}(l+1) = \beta_1 z^{(1)}(l) + \beta_2, \quad l = 1, 2, \dots, m-1 \quad (5)$$

$$\hat{\beta} = (\beta_1, \beta_2)^T = (B^T B)^{-1} B^T Z \quad (6)$$

By taking the transpose of matrix B indicated by B^T in Eq. 6, the value of datum iteration is not considered equal of the value of $z^{(1)}(1)$. To differentiate between both iteration sequences $\hat{z}^{(1)}(1)$ and $z^{(1)}(1)$, a modification is required in β_3 and can be written as $\hat{z}^{(1)}(1) = z^{(1)}(1) + \beta_3$. By taking a least square method, it will assist in solving the complexity of β_3 by using an unconstrained optimized method as follows.

$$\min_{\beta_3} \sum_{m}^{l=1} \left[\hat{z}^{(1)}(l) - z^{(1)}(l) \right]^2 \quad (7)$$

Table 3 Nonhomogeneous exponential sequence X2 with constant term parameters

Model	Exponential sequence	Parameters	Models structuring
NDGM	Non-homogeneous	$\hat{\beta}_1 = (1.7779, 1-0.72994)^T$	$\hat{z}^{(1)}(k+1) = 1.7779z^{(1)}(k) - 0.72994$
ODGM	Non-homogeneous	$\hat{\beta}_2 = (1.8, -1.40, 4.10, 0.0)^T$	$\hat{z}^{(1)}(k+1) = 1.8z^{(1)}(k) - 1.40k + 4.10, \hat{z}^{(1)}(1) = z^{(1)}$
VSSGM	Non-homogeneous	$a_3 = (-0.72997, 0.0, 0.0, -0.89810, 4.0511)^T$	$\hat{z}^{(0)}(k) - 0.72997z^{(1)}(k) = 0.0k^{(3)} + 0.0k^{(2)} - 0.89810k + 4.0511$

Table 4 Nonhomogeneous exponential sequence X3 parameters with velocity and constant terms

Model	Exponential sequence	Parameters	Models structuring
NDGM	Non-homogeneous	$\hat{\beta}_1 = (1.9745, -2.9854)^T$	$\hat{z}^{(1)}(k+1) = 1.9745\hat{z}^{(1)}(k) - 2.9854$
ODGM	Non-homogeneous	$\hat{\beta}_2 = (2.5584, -2.995, 8.0121, -0.4152)^T$	$\hat{z}^{(1)}(k+1) = 2.5584\hat{z}^{(1)}(k) - 2.9950k + 8.0121, \hat{z}^{(1)}(1) = z^{(1)} - 0.4152$
VSSGM	Non-homogeneous	$a_3 = (-0.71254, 0.0, -0.16247, -0.39152, 4.2395)^T$	$\hat{z}^{(0)}(k) - 0.71254\hat{z}^{(1)}(k) = 0.0k^{(3)} - 0.16247k^{(2)} - 0.39152k + 4.2395$

Step 4: Simulated/forecasting values $z^{(1)}(L+1)$ and $z^{(0)}(L+1)$ represent the values of prediction/simulation functions and can be written as follows.

$$\hat{z}^{(1)}(l+1) = \beta_1^l z^{(1)} \left(\hat{z}^{(1)}(1) + \beta_3 \right) + \frac{1-\beta_1^l}{1-\beta_1} \cdot \beta_2, \quad l = 1, 2, \dots, m-1 \quad (8)$$

$$\begin{aligned} \hat{z}^{(0)}(l+1) &= \hat{z}^{(1)}(l+1) - \hat{y}^{(1)}(l) \\ &= (\beta_1^l - \beta_1^{l-1}) \left(\hat{y}^{(1)}(1) + \beta_3 \right) - \frac{\beta_2}{1-\beta_1}, \\ l &= 1, 2, \dots, m-1 \end{aligned} \quad (9)$$

In addition:

$\{\hat{z}^{(0)}(1), \hat{z}^{(0)}(2), \dots, \hat{z}^{(0)}(m)\}$ presented the simulation values, whereas

$\{\hat{z}^{(0)}(m+1), \hat{z}^{(0)}(m+2), \dots, \hat{z}^{(0)}(m+n)\}$ indicated the values of prediction operating sequence.

Step 5: The final step involves the validation and performance accuracy of the ODGM model by using the most adaptable approach, namely, the mean absolute percentage error (MAPE %), which has been used in previous studies (Ikram et al. 2020a).

Nonhomogeneous discrete grey model (NDGM)

The NDGM system is based on non-homogenous exponential growth approximation law in order to comply with the assumptions of an actual data sequence. Wu et al. (2014) recommended that the real data sequence competes with a homogeneous pattern, such as GM (1,1). The accuracy of the NDGM model is significantly improved over the other grey models as far as the mean sequence value, and the value of the intervals is concerned. This NDGM model has been used in various disciplines, such as predicting Turkey's electricity consumption and analyzing that the NDGM best fits and predicts more accurately than other grey predicting models. Duan et al. (2018) predicted the consumption of crude oil in China and analyzed that the NDGM showed superior performance.

In the model of NDGM, $z^{(0)}$ denotes the original data sequence and $z^{(1)}$ represents the accumulated sequence of data. So, they are written as follows.

$$\hat{z}^{(1)}(l+1) = \beta_1 \hat{z}^{(1)}(l) + \beta_2 \cdot l + \beta_3 \quad (10)$$

$$\hat{z}^{(1)}(1) = z^{(1)}(1) + \beta_4 \quad (11)$$

Table 5 Nonhomogeneous exponential sequence X4 parameters with acceleration, velocity, and constant terms

Model	Exponential sequence	Parameters	Models structuring
NDGM	Non-homogeneous	$\hat{\beta}_1 = (1.425, -9.8542)^T$	$\hat{z}^{(1)}(k+1) = 1.425z^{(1)}(k) - 9.8542$
ODGM	Non-homogeneous	$\hat{\beta}_2 = (1.1235, 2.6145, 6.0152, -0.02541)^T$	$\hat{z}^{(1)}(k+1) = 1.12235z^{(1)}(k) + 2.6145k + 6.0152, \hat{z}^{(1)}(1) = z^{(1)} - 0.02541$
VSSGM	Non-homogeneous	$a_3 = (-0.3, -0.07, 0.0, 2.01, 3.12)^T$	$\hat{z}^{(0)}(k) - 0.3z^{(1)}(k) = -0.07k^{(3)} + 0.0k^{(2)} + 2.01k + 3.12$

Then, $\hat{z}^{(1)}(l)$ represents the predicted value of $z^{(1)}$ along with their parameters $\beta_1, \beta_2, \beta_3$ and β_4 . So, the above equation is presented in the matrix form as below, where $L = 1, 2$, and $3 \dots m-1$:

$$\begin{bmatrix} z^{(1)}(2) \\ z^{(1)}(3) \\ \vdots \\ z^{(1)}(m) \end{bmatrix} = \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} \begin{bmatrix} z^{(1)}(1) & 1 & 1 \\ z^{(1)}(2) & 2 & 1 \\ \vdots & \vdots & \vdots \\ z^{(1)}(m-1) & m-1 & 1 \end{bmatrix}$$

The input data in a single case indicates a constant sequence for the NDGM parameters $\beta_1, \beta_2, \beta_3$, and β_4 . So, by applying this relation in Eq. (3), we get it as:

$$\hat{\beta} = (B^T B)^{-1} B^T Z = [\beta_1, \beta_2, \beta_3]^T \quad (12)$$

Whereas,

$$B = \begin{bmatrix} z^{(1)}(1) & 1 & 1 \\ z^{(1)}(2) & 2 & 1 \\ \vdots & \vdots & \vdots \\ z^{(1)}(m-1) & m-1 & 1 \end{bmatrix} Y = \begin{bmatrix} z^{(1)}(2) \\ z^{(1)}(3) \\ \vdots \\ z^{(1)}(m) \end{bmatrix}$$

Further, in order to calculate β_4 , the following formula is used, where we minimized the sum of square error as:

$$\beta_4 = \frac{\sum_{l=1}^{m-1} \left[z^{(1)}(l+1) - \beta_1^l z^{(1)}(1) - \beta_2 \sum_{j=1}^l j \beta_1^{l-j} - \frac{1-\beta_1^l}{1-\beta_1} \cdot \beta_3 \right] \cdot \beta_1^l}{1 - \sum_{l=1}^{m-1} (\beta_1^l)^2} \quad (13)$$

$$\hat{z}^{(1)}(L+1) = \beta_1^L \hat{z}^{(1)}(1) + \beta_2 \sum_{j=1}^L j \beta_1^{L-j} + \frac{1-\beta_1^L}{1-\beta_1} \cdot \beta_3; \quad (14)$$

$L = 1, 2, \dots, m-1$

For the additional insight regarding the NDGM model, its parameter, and properties, see (Liu and Yang 2017).

Variable speed and adaptive structure grey model (VSSGM)

The VSSGM model was proposed by Li et al. (2020a). We used the application of VSSGM for the first time to forecast the future trends of CO₂ emissions and renewable electricity

production of two high emitting countries China and the USA. We have compared VSSGM model with other two novel forecasting models, namely, ODGM and NDGM. This section explains how the VSSGM model is developed and how it provides better accuracy over other forecasting models. The construction of VSSGM model involves four steps.

Step 1: Let suppose that $Z^{(0)}$ is a non-negative sequence of raw data and can be written as

$Z^{(0)} = z^{(0)}(1), z^{(0)}(2), \dots, z^{(0)}(m)$, with α called the sequential operator in which the operation function is satisfying and can be presented as $Z^{(1)} = z^{(1)}(1), z^{(1)}(2), \dots, z^{(1)}(m)$, whereas

$z^{(1)}(l) = \sum_{j=1}^l z^{(0)}(j), l = 1, 2, \dots, m$. According to the defini-

tion of sequential operator, α is known as first-order accumulating generation operator and is presented by 1-AGO. If

AGO accumulating α is applied n times on raw data sequence, which is $Z^{(0)}$, then we can write the equation like:

$Z^{(0)} \alpha^n = Z^{(n)} = (z^{(n)}(1), z^{(n)}(2), \dots, z^{(n)}(m))$, where $z^{(l)} = \sum_{j=1}^l$

$z^{(n-1)}(j), l = 1, 2, \dots, m$. Further, the AGO is used inversely, which is called inverse AGO and can be presented as $\beta^{(n)} z^{(0)}(l) = \alpha^{(n-1)} z^{(0)}(l) - \alpha^{(n-1)} z^{(0)}(l-1)$. The β is also

known as AGO in the above equation. To make the grey model more effective, the randomness of accumulating generation operation of raw sequence data can be attenuated. Conversely, inverse AGO is a mirror of the inverse process of the accumulating generation operation and which has an important significance role in the regression of AGO (Li et al. 2020b).

Step 2: Let suppose that $Z^{(1)}$ is a non-negative sequence of raw data, the same as mentioned in step 1. After, $Z^{(0)}$ a newly generated raw data sequence is determined and can be written as $Z^{(1)} = z^{(1)}(1), z^{(1)}(2), \dots, z^{(1)}(m)$, in which $z^{(1)}(i) = 0.5 * [z^{(1)}(i) + z^{(1)}(i-1)], i = 2, 3, \dots, m$. Whereas, consecutive adjacent elements generated a new data sequence called generation sequence, represented by $Z^{(1)}$ in the (1-AGO) data sequence indicated by $Y^{(1)}$.

Table 6 Random number sequence X5 parameters

Model	Exponential sequence	Parameters	Models structuring
NDGM	Non-homogeneous	$\hat{\beta}_1 = (1.077538, -41.96)^T$	$\hat{z}^{(1)}(k+1) = 1.077538z^{(1)}(k) - 41.96$
ODGM	Non-homogeneous	$\hat{\beta}_2 = (1.00165, 19.5416, 26.2541, 0.16904)^T$	$\hat{z}^{(1)}(k+1) = 1.00165z^{(1)}(k) + 19.5416k + 26.2541, \hat{z}^{(1)}(1) = z^{(1)} + 0.16904$
VSSGM	Non-homogeneous	$\hat{\alpha}_3 = (0.21254, -0.087164, 1.5977, 8.4239, 22.854)^T$	$\hat{z}^{(0)}(k) + 0.21254z^{(1)}(k) = -0.087164k^{(3)} + 1.5977k^{(2)} + 8.4239k + 22.854$

Step 3: If $Z^{(0)}$, $Z^{(1)}$ and $Y^{(1)}$ are derived from steps 1 and 2, then the equation can be written as follows.

$$z^{(0)}(k) + az^{(1)}(k) = b_1k^3 + b_2k^2 + b_3k + b_4, k \in \mathbb{Z}, k \geq 2 \quad (15)$$

$$z^{(0)}(k) = a_0p^k \rightarrow \text{without perturbation}$$

$$z^{(0)}(k) = a_0p^k + a_1 \rightarrow \text{with constant perturbation term}$$

$$z^{(0)}(k) = a_0p^k + a_2k + a_1 \rightarrow \text{with perturbations of constant and velocity}$$

$$z^{(0)}(k) = a_0p^k + a_3k^2 + a_2k + a_1 \rightarrow \text{with disruption of constant, velocity and acceleration}$$

The above equations utilize grey-based system theory and can be called a grey forecasting model with variable speed and structure. Moreover, the development index is presented by -a. Whereas, $b_l (l = 1, 2, 3, 4)$ is known as grey actuating quantity a_0, a_1, a_2, a_3 and p has the constants values, which can be expressed as $a_0 \neq 0, p > 0$. We can say that this is an improved and extended form of the basic grey forecasting model of GM (1,1) model, having the additional accuracy features b_1k^3 , b_2k^2 and b_3k to the actual constant term presented by b_4 . Further, by applying the alterable structuring process in order to obtain a different combination of altering sequences and which are characterized as b_1k^3 , b_2k^2 , b_3k and b_4 . There is a significant feature of the VSSGM model, in which homogeneous exponential sequence and approximate nonhomogeneous exponential sequence with velocity acceleration and the constant term can be simulated simultaneously. Moreover, there is another superior advantage of the VSSGM, and it can change the forecasting structuring according to data nature if $b_1 = 0$. Then, VSSGM can be simulated and downgrade itself as a grey model with velocity and constant term only, which makes the VSSGM model more superior over other forecasting models.

Step 4: If $Z^{(0)}$, $Z^{(1)}$ and $X^{(1)}$ are developed and treated as same as in step 3, then $\hat{a} = (a, b_1, b_2, b_3, b_4)$ is called the estimated parameter vector, presented as follows:

$$Y = \begin{bmatrix} z^{(0)}(2) \\ z^{(0)}(3) \\ \vdots \\ z^{(0)}(m) \end{bmatrix}, B = \begin{bmatrix} -z^{(1)}(2) & 2^3 & 2^2 & 2 & 1 \\ -z^{(1)}(3) & 3^3 & 3^2 & 3 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ -z^{(1)}(m) & m^3 & m^2 & m & 1 \end{bmatrix}$$

Then, if the least square estimation of the VSSGM model is satisfied, it can be written as follows:

$$\hat{a} = (a, b_1, b_2, b_3, b_4)^T = (B^T B)^{-1} B^T Y. \quad (16)$$

Step 5: In the final step of VSSGM construction, $\hat{z}^{(1)}(t)$ describes the time response sequence and $\hat{z}^{(0)}(t)$

Table 7 China's electricity production simulation and prediction values and errors of three models based on random sequence number

Year	Original Value of electricity generation (Billion KWh)	ODGM ($\beta_1 = 1.1111$, $\beta_2 = 2.4444$)		NDGM ($\beta_1 = 1.1111$, $\beta_2 = 1.1$ 2.4444, $\beta_3 = 1.1111$, $\beta_4 = 1.1111$)		VSSGM (a = 1211, $\beta_1 = 1.1111$, $\beta_2 = 1.1$ 2.4444, $\beta_3 = 1.1111$, $\beta_4 = 1.1111$)	
		simulation value $\hat{z}^{(0)}(k)$	simulation error $\Delta_k(\%)$	simulation value $\hat{z}^{(0)}(k)$	simulation error $\Delta_k(\%)$	simulation value $\hat{z}^{(0)}(k)$	simulation error $\Delta_k(\%)$
2000	3.056	3.056	0	3.056	0	3.056	0
2001	3.211	4.08526655	1.097831	4.843111	1.0455	3.72547	1.0151
2002	3.343	7.345358475	2.293862	6.9955795	2.1846	5.381215	2.1210
2003	3.524	5.68291542	1.429094	5.4123004	1.3611	4.163308	1.3214
2004	3.814	1.578599295	2.87506	1.5034279	2.7382	1.156483	2.6584
2005	4.623	9.37210638	2.867489	8.9258156	2.7309	6.866012	2.6514
2006	6.474	22.28817045	9.966455	21.226829	9.4919	16.32833	9.2154
2007	8.318	36.85684275	2.798273	35.101755	2.6650	27.00135	2.5874
2008	29.811	55.2314217	5.639806	52.601354	5.3712	40.46258	5.2148
2009	48.145	78.406146	4.558955	74.67252	4.3418	57.4404	4.2154
2010	79.354	107.6350412	1.314455	102.509563	1.2518	78.85351	1.2154
2011	111.342	144.499719	5.296106	137.61878	5.0439	105.8606	4.8970
2012	143.436	190.9947585	4.76482	181.89977	3.1093	139.9229	2.145
2013	206.113	274.2062505	5.675496	261.14881	5.40523	200.8837	5.2478
2014	242.799	323.5971375	4.244347	308.18775	4.04223	237.0675	3.9245
2015	294.441	416.8795995	7.92361	397.02819	7.54629	305.4063	7.3265
2016	388.604	534.5311335	2.934326	509.07727	2.79459	391.5979	2.7132
$\overline{\Delta}(\text{Simulation})$			5.9799		5.6952		5.5294
		Predictive value $\hat{z}^{(0)}(k)$	Predictive error $\Delta_k(\%)$	Predictive value $\hat{z}^{(0)}(k)$	Predictive error $\Delta_k(\%)$	Predictive value $\hat{z}^{(0)}(k)$	Predictive error $\Delta_k(\%)$
2017		682.91769	15.639429	650.3978	12.03033	500.306	9.2541
2018		870.0688815	10.547628	828.63703	8.11356	637.4131	6.2412
2019		1106.111234	12.192505	1053.43927	9.37885	810.3379	7.2145
2020		1403.816505	16.920956	1336.9681	13.01612	1028.437	10.0124
2021		1779.295245	10.176335	1694.5669	7.82795	1303.513	6.0215
2022		2252.862885	10.547628	2145.5837	8.11356	1650.449	6.2412
2023		2850.145935	15.572843	2714.4247	11.97911	2088.019	9.2147
2024		3603.462135	8.814026	3431.8687	6.78002	2639.899	5.2154
2025		4553.575845	12.405445	4336.7389	9.54265	3335.953	7.3405
2026		5751.89433	13.884026	5477.9946	10.68002	4213.842	8.2154
$\overline{\Delta}(\text{Forecasting})$			12.6700		9.7462		7.4173

expresses the regressive reduction sequence, which is obtained from Eq. (7) and cannot be utilized for forecasting operation directly. First, there is a need to deduct the time response sequence, written as $\hat{z}^{(1)} = \{\hat{z}^{(1)}(k), k \in \mathbb{N}, k \geq 1\}$, and regressive reduction corresponding sequence $\hat{z}^{(0)} = \{\hat{z}^{(0)}(k), k \in \mathbb{N}, k \geq 1\}$. According to the estimation parameters $\hat{a} = (a, b_1, b_2, b_3, b_4)^T$, the

sequence of time response $\hat{z}^{(1)}(k)$ and the reductive regressive sequence, which $\hat{z}^{(0)}(k)$, can be written as follows:

$$\begin{cases} \hat{z}^{(1)}(k) = m^{k-1} \hat{z}^{(1)}(1) + n \sum_{j=1}^4 b_j \sum_{l=2}^k m^{k-l} t^{4-j}, k \in \mathbb{N}, k \geq 2, \\ \hat{z}^{(1)}(1) = z^{(1)}(1); \end{cases} \quad (17)$$

Table 8 The US electricity production simulation and prediction values & errors of three models based on random sequence number

Year	Original Value of electricity generation (Billion Kwh)	ODGM ($\beta_1 = 1.1111$, $\beta_2 = 2.4444$)		NDGM ($\beta_1 = 1.1111$, $\beta_2 = 1.1$ 2.4444, $\beta_3 = 1.1111$, $\beta_4 = 1.1111$)		VSSGM ($a = 1211$, $\beta_1 = 1.1111$, $\beta_2 = 1.1$ 2.4444, $\beta_3 = 1.1111$, $\beta_4 = 1.1111$)	
		simulation value $\hat{z}^{(0)}(k)$	simulation error $\Delta_k(\%)$	simulation value $\hat{z}^{(0)}(k)$	simulation error $\Delta_k(\%)$	simulation value $\hat{z}^{(0)}(k)$	simulation error $\Delta_k(\%)$
2000	81.012591	81.01259	0	81.01259	0	81.01259	0
2001	82.80389	113.601074	2.001552	108.191499	1.42968	83.22423	1.0212
2002	92.7888716	125.0961075	1.945104	119.13915	1.38936	91.6455	0.9924
2003	93.7097026	125.6157767	3.002916	119.634073	2.14494	92.02621	1.5321
2004	97.5207425	133.1007542	1.670704	126.762623	1.19336	97.50971	0.8524
2005	100.482027	146.410992	6.386464	139.43904	4.56176	107.2608	3.2584
2006	109.983481	161.709912	7.575792	154.00944	5.41128	118.4688	3.8652
2007	118.151398	179.2947975	7.881944	170.75695	5.62996	131.3515	4.0214
2008	139.001983	199.5071715	4.549944	190.00683	3.24996	146.1591	2.3214
2009	157.681933	222.739608	5.849228	212.13296	4.17802	163.1792	2.9843
2010	182.357073	249.443376	0.44002	237.56512	0.3143	182.7424	0.2245
2011	211.826314	280.137039	6.517784	266.79718	4.65556	205.2286	3.3254
2012	238.049096	315.416829	5.851384	300.39698	4.17956	231.0746	2.9854
2013	275.227052	355.967976	6.955452	339.01712	4.96818	260.7824	3.5487
2014	303.906652	402.578085	6.942516	383.4077	4.95894	294.9291	3.5421
2015	323.327979	456.1525605	4.98134	434.43101	3.5581	334.1777	2.5415
2016	374.199554	517.731942	8.040116	493.07804	5.74294	379.2908	4.1021
$\bar{\Delta}(\text{Simulation})$			5.03701625		3.59786875		2.5699
		Predictive value $\hat{z}^{(0)}(k)$	Predictive error $\Delta_k(\%)$	Predictive value $\hat{z}^{(0)}(k)$	Predictive error $\Delta_k(\%)$	Predictive value $\hat{z}^{(0)}(k)$	Predictive error $\Delta_k(\%)$
2017		588.512106	5.32145	560.48772	4.52134	431.1444	2.1403
2018		669.8681535	7.198386	637.96967	5.53722	490.7459	4.2594
2019		763.380072	4.296149	727.02864	3.30473	559.2528	2.5421
2020		870.8641305	12.194026	829.39441	9.38002	637.9957	7.2154
2021		994.40796	8.814533	947.0552	6.78041	728.504	5.2157
2022		1136.411231	13.884026	1082.29641	10.68002	832.5357	8.2154
2023		1299.632198	7.124026	1237.74495	5.48002	952.1115	4.2154
2024		1487.24121	10.531235	1416.4202	8.10095	1089.554	6.2315
2025		1702.88118	13.884533	1621.7916	10.68041	1247.532	8.2157
2026		1950.741975	17.297826	1857.8495	13.30602	1429.115	10.2354
$\bar{\Delta}(\text{Forecasting})$			11.0985		8.5373		6.5671

and

When $k = 1$, the conclusion can be drawn. If $k = 2$ is written as:

$$\begin{cases} \hat{z}^{(0)}(k) = \hat{z}^{(1)}(k) - \hat{z}^{(1)}(k-1), k \in \mathbb{N}, k \geq 2, \\ \hat{z}^{(0)}(1) = z^{(0)}(1). \end{cases} \quad (18)$$

$$\hat{z}^{(1)}(2) = m\hat{z}^{(1)}(1) + n(b_1 2^3 + b_2 2^2 + b_3 2 + b_4) \quad (19)$$

$$= m^{2-1}\hat{z}^{(1)}(1) + n \sum_{j=1}^4 b_j \sum_{i=2}^2 m^{2-i} t^{4-j} \quad (20)$$

where $m = \frac{1-0.5a}{1+0.5a}$, $n = \frac{1}{1+0.5a}$ and $\hat{a} = (a, b_1, b_2, b_3, b_4)^T$ are estimated.

then, (2) when the conclusion is correct in any conditions, and when $k = p \in \mathbb{N}$ and $p > 2$, then the following sequence of the equation can be written as follows:

Table 9 China's CO₂ emissions simulation and prediction values and errors of three models based on random sequence number

Year	Original Value of CO ₂ emissions (mmTons)	ODGM ($\beta_1 = 1.1111$, $\beta_2 = 2.4444$)		NDGM ($\beta_1 = 1.1111$, $\beta_2 = 1.1$ 2.4444, $\beta_3 = 1.1111$, $\beta_4 = 1.1111$)		VSSGM (a = 1211, $\beta_1 = 1.1111$, $\beta_2 = 1.1$ 2.4444, $\beta_3 = 1.1111$, $\beta_4 = 1.1111$)	
		<u>simulation</u> value $\hat{z}^{(0)}(k)$	<u>simulation</u> error $\Delta_k(\%)$	<u>simulation</u> value $\hat{z}^{(0)}(k)$	<u>simulation</u> error $\Delta_k(\%)$	<u>simulation</u> value $\hat{z}^{(0)}(k)$	<u>simulation</u> error $\Delta_k(\%)$
2000	3523.15227	3523.15227	0	3523.1523	0	3523.152	0
2001	3694.86799	4543.494806	2.001944	3803.328387	1.42996	3701.868	1.0214
2002	3960.51443	5406.102197	3.906672	4048.668759	2.79048	3742.31	1.9932
2003	4625.45309	6313.743468	0.024304	4513.089017	0.01736	4608.054	0.0124
2004	5375.48102	7337.531592	2.37944	5988.125326	1.6996	5401.819	1.2140
2005	6105.59822	8334.14157	1.217944	6937.277686	0.86996	6129.59	0.6214
2006	6739.96121	9200.047052	0.298116	7761.949573	0.21294	6796.854	0.1521
2007	7218.93073	9853.840446	4.340616	8384.609949	3.10044	7408.641	2.2146
2008	7496.67722	10,232.96441	9.732772	9045.680386	6.95198	7969.563	4.9657
2009	8188.76519	11,177.66448	8.360968	9945.39475	5.97212	8483.85	4.2658
2010	8779.18976	11,983.59402	4.608744	11,412.94669	3.29196	8955.378	2.3514
2011	9832.27819	13,421.05973	13.038704	12,781.96165	9.31336	9387.704	6.6524
2012	10,362.2039	14,144.40832	10.43994	13,470.86507	7.4571	9625.201	5.3265
2013	10,801.7734	14,744.42069	10.348408	14,042.30542	7.39172	10,147.51	5.2798
2014	10,744.2471	14,665.89729	4.250652	13,967.52123	3.03618	10,480.72	2.1687
2015	10,603.674	14,474.01501	5.634804	13,784.7762	4.02486	10,786.23	2.8749
2016	10,297.7967	14,056.4925	8.360968	13,387.13571	5.97212	11,066.34	4.2658
$\overline{\Delta}(\text{Simulation})$			2.8363		3.9708		2.5362
		<u>Predictive</u> value $\hat{z}^{(0)}(k)$	<u>Predictive</u> error $\Delta_k(\%)$	<u>Predictive</u> value $\hat{z}^{(0)}(k)$	<u>Predictive</u> error $\Delta_k(\%)$	<u>Predictive</u> value $\hat{z}^{(0)}(k)$	<u>Predictive</u> error $\Delta_k(\%)$
2017		15,456.1	7.750678	14720.1	5.96206	11,323.15	4.5862
2018		15,777.52	8.166249	15,026.21	6.28173	11,558.62	4.8321
2019		16,072.21	9.574526	15,306.86	7.36502	11,774.51	5.6654
2020		16,342.39	8.99756	15,564.19	6.9212	11,972.45	5.3240
2021		16,590.11	7.157826	15,800.11	5.50602	12,153.93	4.2354
2022		16,817.25	6.757465	16,016.43	5.19805	12,320.33	3.9985
2023		17,025.49	8.993166	16,214.76	6.91782	12,472.89	5.3214
2024		17,216.43	8.999926	16,396.6	6.92302	12,612.77	5.3254
2025		17,391.48	8.438508	16,563.31	6.49116	12,741.01	4.9932
2026		17,551.99	9.224189	16,716.18	7.09553	12,858.6	5.4581
$\overline{\Delta}(\text{Forecasting})$			8.4060		6.4662		4.9740

$$\hat{z}^{(1)}(p) = m^{p-1}\hat{z}^{(1)}(1) + n \sum_{j=1}^4 b_j \sum_{l=2}^p m^{p-l} l^{4-j} \quad (21)$$

$$= m^p \hat{z}^{(1)}(1) + n \left(b_1 \sum_{l=2}^p m^{p+1-l} l^3 + b_2 \sum_{l=2}^p m^{p+1-l} l^2 + b_3 \sum_{l=2}^p m^{p+1-l} l + b_4 \sum_{l=2}^p m^{p+1-l} \right) + n \left(b_1(p+1)^3 + b_2(p+1)^2 + b_3(p+1) + b_4 \right) \quad (24)$$

If $k = p + 1$, then it is presented as follows:

$$\hat{z}^{(1)}(p+1) = m \hat{z}^{(1)}(p) + n \left(b_1(p+1)^3 + b_2(p+1)^2 + b_3(p+1) + b_4 \right) \quad (22)$$

$$= m \left(m^{p-1} \hat{z}^{(1)}(1) + n \sum_{j=1}^4 b_j \sum_{l=2}^p m^{p-l} l^{4-j} b_j \right) + n \left(b_1(p+1)^3 + b_2(p+1)^2 + b_3(p+1) + b_4 \right) \quad (23)$$

$$= m^p \hat{z}^{(1)}(1) + n \sum_{j=1}^4 b_j \sum_{l=2}^{p+1} m^{p+1-l} l^{4-j} \quad (26)$$

Therefore, $\hat{Z}(1)$ is the time response sequence, where-
as $\hat{Z}(0)$ is the regressive reduction sequence and

Table 10 The USA's CO₂ emissions simulation and prediction values and errors of three models based on random sequence number

Year	Original Value of CO ₂ emissions (mmTons)	ODGM ($\beta_1 = 1.1111$, $\beta_2 = 2.4444$)		NDGM ($\beta_1 = 1.1111$, $\beta_2 = 1.1$ 2.4444, $\beta_3 = 1.1111$, $\beta_4 = 1.1111$)		VSSGM ($a = 1211$, $\beta_1 = 1.1111$, $\beta_2 = 1.1$ 2.4444, $\beta_3 = 1.1111$, $\beta_4 = 1.1111$)	
		<u>simulation value</u> $\hat{z}^{(0)}(k)$	<u>simulation error</u> $\Delta_k(\%)$	<u>simulation value</u> $\hat{z}^{(0)}(k)$	<u>simulation error</u> $\Delta_k(\%)$	<u>simulation value</u> $\hat{z}^{(0)}(k)$	<u>simulation error</u> $\Delta_k(\%)$
2000	8034.64834	8034.648	0	8034.648	0	8034.648	0
2001	7857.59618	9270.66354	2.2197	8225.7748	1.5855	7890.596	1.1325
2002	7904.99004	11,399.00171	8.300796	9656.1921	5.92914	8350.917	4.2351
2003	7994.55215	9703.09954	6.302184	8764.8567	4.50156	8280.659	3.2154
2004	8139.82403	9690.52799	4.557	8657.6457	3.255	8198.189	2.3250
2005	8183.48932	9558.39053	2.598568	8531.8005	1.85612	8101.385	1.3258
2006	8075.5368	9503.28694	2.597784	8538.0828	1.85556	7987.756	1.3254
2007	8201.95287	9121.22324	11.08184	8210.6888	7.9156	7854.376	5.6540
2008	7990.25333	9007.51475	6.400184	8107.1569	4.57156	7697.813	3.2654
2009	7319.03078	9056.66187	7.026404	8068.2494	5.01886	7514.038	3.5849
2010	7579.47733	8962.2068	7.8302	7887.816	5.593	7298.32	3.9950
2011	7479.40495	8816.57242	8.477784	7758.6404	6.05556	7045.108	4.3254
2012	7061.14285	8510.86166	7.026012	7572.2492	5.01858	6747.884	3.5847
2013	5512.01467	8134.633635	12.450116	7018.6987	8.89294	6398.999	6.3521
2014	5555.17602	7775.63201	14.35994	6686.3162	10.2571	5989.474	7.3265
2015	5414.63008	7219.46832	2.420796	6061.3984	1.72914	5508.768	1.2351
2016	5316.90929	6749.254785	11.555572	5832.5483	8.25398	4944.509	5.8957
$\overline{\Delta}(\text{Simulation})$			7.2003		5.1431		3.6736
		<u>Predictive value</u> $\hat{z}^{(0)}(k)$	<u>Predictive error</u> $\Delta_k(\%)$	<u>Predictive value</u> $\hat{z}^{(0)}(k)$	<u>Predictive error</u> $\Delta_k(\%)$	<u>Predictive value</u> $\hat{z}^{(0)}(k)$	<u>Predictive error</u> $\Delta_k(\%)$
2017		2847.646375	8.997898	2997.5225	6.92146	4282.175	5.3242
2018		2330.638135	5.509062	2453.3033	4.23774	3504.719	3.2598
2019		1723.76778	3.923335	1814.4924	3.01795	2592.132	2.3215
2020		1011.41579	9.068033	1064.6482	6.97541	1520.926	5.3657
2021		840.248115	14.077362	884.4717	10.82874	1263.531	8.3298
2022		739.75531	12.379419	778.6898	9.52263	1112.414	7.3251
2023		694.85717	3.744026	731.4286	2.88002	1044.898	2.2154
2024		687.28282	2.241785	723.4556	1.72445	1033.508	1.3265
2025		642.1143575	14.071616	675.90985	10.82432	965.5855	8.3264
2026		596.216656	12.269062	627.59648	9.43774	896.5664	7.2598
$\overline{\Delta}(\text{Forecasting})$			8.6282		6.6371		5.1054

established automatically because it is deduced by $\hat{X}(1)$, as shown in Fig. 2.

After the successful step-by-step development of all three grey prediction models based on their properties and structure, the VSSGM model is a more accurate model than others. The VSSGM model allows the unification of the traditional grey prediction model with the homogeneous/non-homogenous exponential models and single variable linear autoregressive model. We employed

three novel grey models to predict renewable electricity production and CO₂ emission for China and the USA. MAPE % method was applied to check the performance accuracy of all three predicting models, which has widely used in several studies (Ikram et al. 2020a; Javed and Liu 2018; Pirthee 2017; Javed et al. 2020), by measuring ODGM, NDGM, and VSSGM criteria between CO₂ and production of renewable electricity, and can be written as follows:

$$MAPE(\%) = \frac{1}{m} \sum_{k=1}^m \left| \frac{z^{(0)}(k) - \hat{z}^{(0)}(k)}{z^{(0)}(k)} \right| \quad (27)$$

Here, $z^{(0)}(k)$ and $\hat{z}^{(0)}(k)$ in Eq. (27) are expressing a model based on the original sequence and forecasted sequence data values. MAPE % represents the performance criteria for comparative analysis between three grey models (Kim and Kim 2016). This study adopted the three novel grey forecasting models to estimate the CO₂ emissions and production of electricity production in 2026 and is considered for the USA and China. The comparison criteria of the MAPE % accuracy level are in Table 1.

The growth of electricity production, CO₂ emissions, and doubling time analysis

There is no model that considers both CO₂ emissions and renewable electricity generation for monitoring growth rate. For example, the RGR model was applied to analyze the US and China CO₂ emissions and electricity production relative growth. Then, the literature underlined the need for considering relative growth rate and doubling time models, which is applied in ISO 14001 certification and research output growth over time (Ikram et al. 2019; Javed and Liu 2018).

Using the NDGM method, two parameters (D_t and RGR) were used to forecast CO₂ emissions of China and the USA. The RGR formula is given by:

$$RGR = (\ln N_2 - \ln N_1) / (t_2 - t_1) \quad (28)$$

where N_2 depicts cumulative electricity production and CO₂ emissions in the year and t_2 and t_1 signify CO₂ emissions in the year t_1 . In the analysis, as $t_2 - t_1$ is 1 year, Eq. (28) can be interpreted as Eq. (29).

$$RGR = \ln(N_2/N_1) \quad (29)$$

Although doubling time (D_t) depicts the time required to reduce CO₂ emissions, double in numbers concerning specified RGR, so, the (D_t) is presented as:

$$D_t = (t_2 - t_1) \ln[2 / (\ln N_2 - \ln N_1)] \quad (30)$$

We can also rewrite this equation as follows:

$$D_t = \ln[2 / (\ln N_2 - \ln N_1)] \quad (31)$$

If the RGR and Doubling Time give a different pattern for actual and simulated data. In this case, the synthetic RGR and Synthetic Doubling Time models used to solve this issue. The synthetic relative growth rate (RGR_{syn}) model is characterized in the following equation as:

$$D_{syn} = \theta(D_a) + (1 - \theta)(D_f) \quad (32)$$

whereas RGR_a is a relative growth rate of actual values of data, while RGR_f depicts forecasted values of the relative growth rate. However, θ represents the relative weight coefficient and is considered to be 0.5.

Then, synthetic doubling time (D_{syn}) model is shown below in Eq. (33).

$$D_{syn} = \theta(D_a) + (1 - \theta)(D_f) \quad (33)$$

where D_a signifies the doubling time acquired from actual values, whereas RGR_f represents the relative growth Rate of forecasted values.

Results and discussion

Our research framework tested three novel grey models (ODGM, NDGM, and VSSGM) for forecasting the electricity generated by renewable sources and CO₂ emissions by

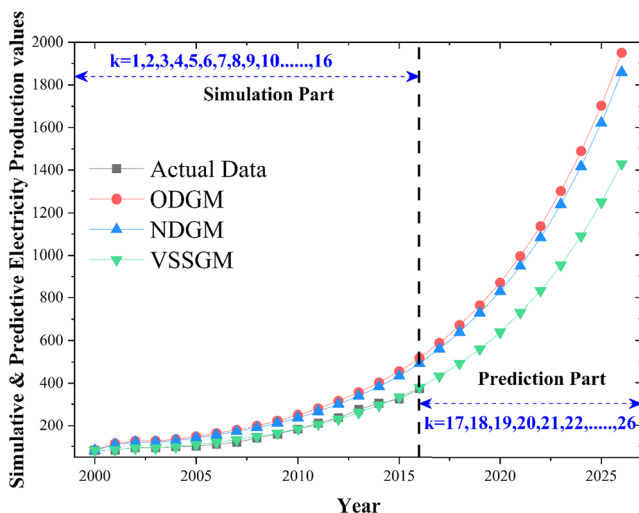


Fig. 5 The US simulation and predictive values of electricity production from 2000 to 2026

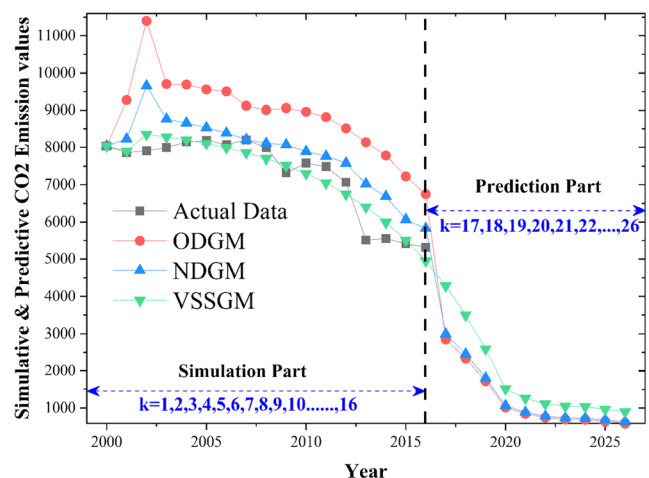


Fig. 6 The USA's simulation and predictive values of CO₂ emission from 2000 to 2026

Table 11 MAPE % by ODGM, NDGM, and VSSGM

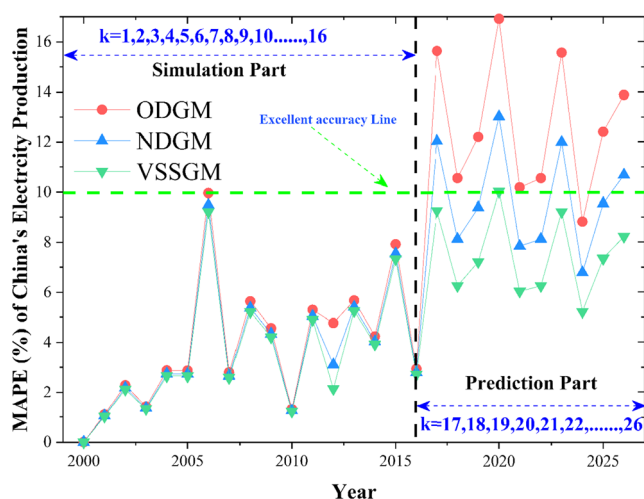
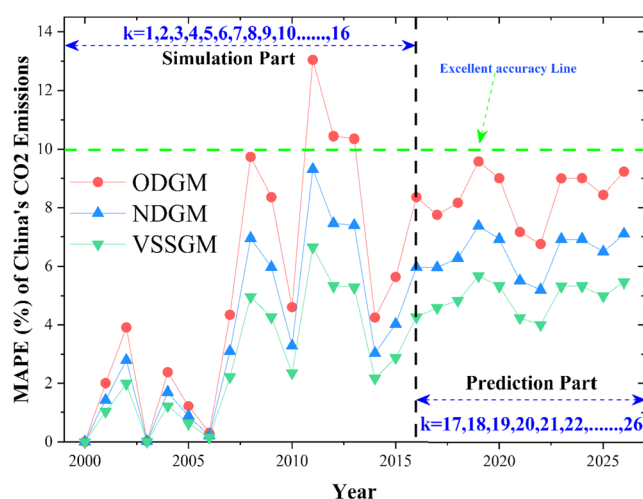
	MAPE %					
	Electricity production			CO ₂ Emissions		
	ODGM	NDGM	VSSGM	ODGM	NDGM	VSSGM
China	9.32	7.72	4.57	5.62	5.22	3.75
USA	8.07	6.07	4.56	7.93	5.89	4.39
Average MAPE %	8.695	6.895	4.565	6.775	5.555	4.07
Overall accuracy level %	91.305	93.105	95.435	93.225	94.445	95.93

comparing the USA and China data out to 2026. These tests used an available dataset comprising the period of 2000–2016. Simulations used the mathematical grey forecasting models for assessing values are presented in Tables 7, 8, 9, and 10. According to the ODGM model, China's electricity production by renewable sources has increased from 3.056 to 534.5311 (Billion KWh) for the period 2000 to 2016. Meanwhile, in the case of China, NDGM calculated that electricity production by renewable sources has also shown an increasing trend from 3.056 to 509.07727 (Billion Kwh) in the same period. Whereas, the VSSGM simulation model values showed exciting results (Table 7) and represents the results for China's electricity production for the period of 2000 to 2016.

The results reveal that China's electricity production by renewable sources has increased from 3.056 to 391.5979 (Billion KWh) for the period 2000–2016. It has observed in our analysis that VSSGM contains smaller simulated errors than other models, as shown in Table 7. By comparison, we found that the VSSGM has the best prediction performance over ODGM and NDGM models. Further, China's electricity forecasting values obtained from 2017 to 2026 by the three prediction

models are in Table 7. Less gap was observed between forecasting values of ODGM (5751.89433; Billion KWh) and NDGM (5477.9946; Billion KWh) on 2026, as can be seen in Fig. 3. Whereas, the VSSGM forecasting value recorded for China's electricity production by renewable sources in 2026 is 4213.842; Billion Kwh. As shown in Table 7, VSSGM has a smaller predictive error value in %, which is 7.42 %, followed by NDGM (9.75%) and ODGM (12.67%), respectively.

CO₂ emissions showed an increasing trend at the same time, as presented in Fig. 4. China's CO₂ emission values calculated by ODGM, NDM, and VSSGM are in Table 9. According to the ODGM model, China's CO₂ has increased from 3523.15 to 14,056.49 (mm tons) for the period 2000 to 2016. Meanwhile, in NDGM simulation, the CO₂ emissions value for China also showed an increasing trend from 3523.15 to 13,387.13571 (mm tons) in the same period of time. Whereas, the VSSGM model calculated the simulation values for China's CO₂ emissions from 3523.15 to 11066.34 for the period of 2000 to 2016. For any given random sequence X5, VSSGM has 2.54% simulation errors, less than ODGM (2.83%) NDGM (3.97%) scores, as shown in Table 9. By

**Fig. 7** China's simulation and predictive MAPE (%) of electricity production from 2000 to 2026**Fig. 8** China's simulation and predictive MAPE (%) of CO₂ emissions from 2000 to 2026

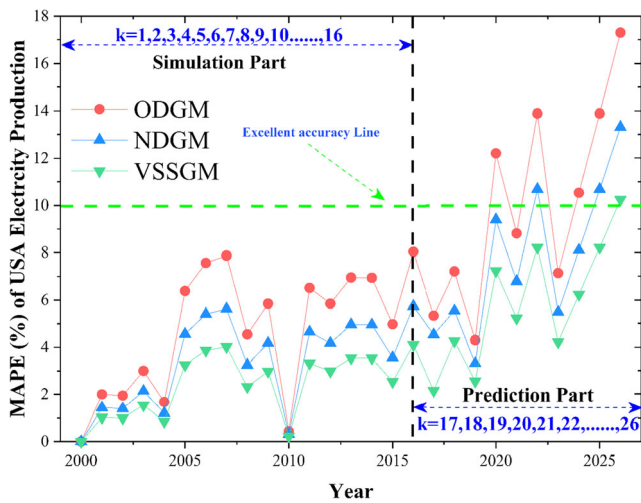


Fig. 9 The US simulation and predictive MAPE (%) of electricity production from 2000 to 2026

comparison, we found that VSSGM and ODGM have the best prediction performances.

China's CO₂ emissions forecasting values from 2017 to 2026 were analyzed through three prediction models with the results in Table 9. All three prediction models showed the inclined forecasting trends of China's CO₂ emissions. There is a slightly less difference observed between forecasting values of China CO₂ emissions obtained by ODGM (17551.99 mm tons) and NDGM (16716.18 mm tons) in 2026. Whereas, the VSSGM forecasting value recorded for China's electricity production by renewable sources, in 2026, is 12,858.6 mm tons. As shown in Fig. 4, VSSGM has shown better prediction performance than NDGM and ODGM.

First, ODGM, NDGM and VSSGM models were used by taking first seventeen original data sequences into consideration for prediction results ($k = 1; 2; \dots; 17$). Taking this to a step further, ODGM, NDGM, and VSSGM were utilized to

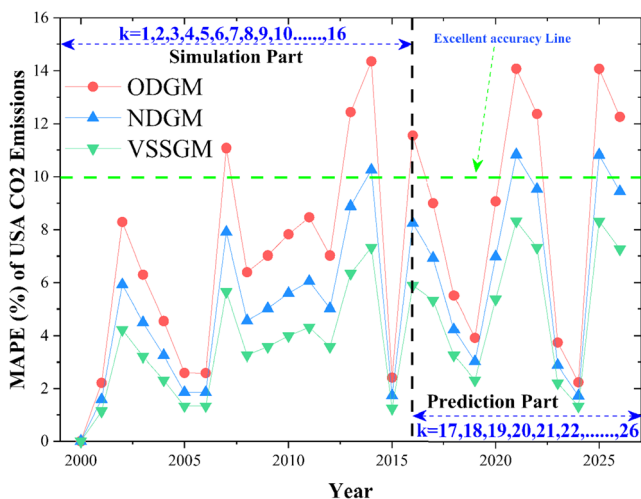


Fig. 10 The US simulation and predictive MAPE (%) of CO₂ emissions from 2000 to 2026

compute the 10 years prediction result, such as $k = 18; 12; 13; 14; 15; \dots; 26$. All the three-model simulations and forecasting performances were assessed and comparatively analyzed. We provide the modeling procedure followed by the analysis and prediction among these models.

The first seventeen numbers of $Z^{(1)}$ based on the 1-AGO $Z^{(1)}(1)$ are analyzed, whereas the mean sequence $Z^{(1)}(1)$ for different models. The estimating parameters are in Table 2.

The first seventeen numbers of X_2 based on the 1-AGO $X^{(1)}(2)$ are analyzed, whereas the mean sequence $Z^{(1)}(2)$ for ODGM, NDGM, and VSSGM and estimating parameters can be seen in Table 3. All the three-model simulation values, forecasting values, and the relative corresponding error values are in Tables 7, 8, 9, and 10.

The first seventeen numbers of X_3 based on the 1-AGO $X^{(1)}(3)$ are analyzed, whereas the mean sequence $Z^{(1)}(3)$ for ODGM, NDGM, and VSSGM models and estimating parameters are in Table 4. All the three models simulated values, forecasting values, and the relative corresponding error values are in Tables 7, 8, 9, and 10.

Similarly, the first seventeen numbers of X_4 based on the 1-AGO $X^{(1)}(4)$ are analyzed, whereas the mean sequence $Z^{(1)}(4)$ for ODGM, NDGM, and VSSGM models and estimating parameters are in Table 5.

Finally, the first seventeen numbers of X_5 based on the 1-AGO $X^{(1)}(5)$ are analyzed, whereas the mean sequence $Z^{(1)}(5)$ for ODGM, NDGM, and VSSGM models, and estimating parameters are in Table 6. All the three models simulation and forecasting values and the relative corresponding error values are in Tables 7, 8, 9, and 10.

According to VSSGM simulation model values, exciting results can be found in Table 8 and represent the results for the US electricity production for the period of 2000–2016 based on the original data. Results revealed that the US electricity production by renewable sources has increased from 81.012 to 379.2908 (Billion KWh) for the period 2000–2016. NDGM calculated that electricity production by renewable sources has also shown the increasing trend from 81.012 to 493.07804 (Billion KWh) in the same period. Whereas, ODGM values showed that the US electricity production by renewable sources has increased from 81.012 to 517.731942 (Billion KWh) for the period 2000–2016.

Meanwhile, in the case of the USA, NDGM and ODGM have more significant simulation errors than VSSGM, as shown in Fig. 5. We found that VSSGM is the better predictor if compared with ODGM and NDGM. Further, the US electricity forecasting values obtained from 2017 to 2026 by three prediction models are in Table 8. Less of a gap was observed between forecasting values of ODGM (1950.74197 Billion KWh) and NDGM (1857.8495 Billion KWh) in 2026. Whereas, the VSSGM forecasting value recorded for the US electricity production by renewable sources in 2026 is 1429.115 Billion KWh. As shown in Table 8, VSSGM has a

Table 12 China relative growth rate and doubling time model results of electricity production

Years	Aggregate values	Relative growth rate (RGR)	Mean RGR	Doubling time (D_t)	Mean D_t
2000	3.056				
2001	6.267	0.718	0.677	1.024	2.813
2002	9.61	0.428		1.543	
2003	13.134	0.312		1.857	
2004	16.948	0.255		2.060	
2005	21.571	0.241		2.115	
2006	28.045	0.262		2.031	
2007	36.363	0.260		2.041	
2008	66.174	0.599		1.206	
2009	114.319	0.547		1.297	
2010	193.673	0.527		1.333	
2011	305.015	0.454		1.482	
2012	152432	6.214		-1.134	
2013	152,614.88	0.001		7.419	
2014	152,851.95	0.002		7.161	
2015	153,157.36	0.002		6.910	
2016	153,548.96	0.003		6.663	
2017	500.306		0.399		1.682
2018	1137.7191	0.822		0.890	
2019	1948.0569	0.538		1.313	
2020	2976.4943	0.424		1.551	
2021	4280.0075	0.363		1.706	
2022	5930.457	0.326		1.814	
2023	8018.4763	0.302		1.892	
2024	10,658.376	0.285		1.950	
2025	13,994.328	0.272		1.994	
2026	18,208.17	0.263		2.028	

smaller predictive error value, which is 6.57 %, followed by NDGM (8.54%) and ODGM (11.10%), respectively.

The US CO₂ emissions showed a decreasing trend for the same period, as presented in Fig. 6. In the USA, CO₂ emissions values calculated by ODGM, NDM, and VSSGM are in Table 10. VSSGM model simulation values have observed the declining trend for the US CO₂ emissions from 8034.64834 to 4944.509 (mm tons), considering 2000 up to 2016. Meanwhile, the ODGM identified that CO₂ emissions by the USA have decreased from 8034.64834 to 6749.254785 (mm tons) for the same period.

Similarly, NDGM simulation on CO₂ emissions values for the USA also showed an increasing trend from 8034.64834 to 5832.5483 (mm tons) for the same period. Interestingly, in the US CO₂ simulation case, VSSGM showed better performance with smaller simulation errors (3.67%), followed by NDGM (5.14%) and ODGM (7.20%), as shown in Table 10. Similarly, Fig. 6 indicates, in the case of the USA, that CO₂ emissions decreased to 61% in mm tons; meanwhile, electricity generation through renewable sources has increased from

81.012591 to 374.199554 in Billion KWh, as presented in Table 8.

The US CO₂ emissions forecasting values obtained from 2017 to 2026 by the three prediction models are in Table 10. All the three prediction models showed the declined forecasting trends of the US CO₂ emissions, as presented in Fig. 6. There is a small gap observed between forecasting values of the US CO₂ emissions obtained by ODGM (596.216656 mm tons) and NDGM (627.59648 mm tons) in 2026. Whereas, the VSSGM forecasting value recorded for the US electricity production by renewable sources in 2026 is 896.5664 mm tons. As shown in Table 10, VSSGM has better prediction performance than NDGM and ODGM.

The ODGM, NDGM, and VSSGM showed overall accuracy and effectiveness obtained by mean absolute percentage error (MAPE %) at 91.305, 93.105, and 95.435%, respectively, of the electricity production, and 93.225, 94.445, and 95.93 of the CO₂ emissions for China and the USA, as presented in Table 11 and can be seen in Figs. 7, 8, 9, and 10. The accuracy of VSSGM is higher than NDGM and ODGM. MAPE % also

Table 13 The US relative growth rate and doubling time model results of CO₂ emissions

Years	Aggregate values	Relative growth rate (RGR)	Mean RGR	Doubling time (Dt)	Mean Dt
2000	8034.648				
2001	15,892.24	0.682068	0.170841	1.075773	2.79777
2002	23,797.23	0.403738		1.600136	
2003	31,791.79	0.289639		1.932269	
2004	39,931.61	0.22796		2.171731	
2005	48,115.1	0.186428		2.372858	
2006	56,190.64	0.155154		2.556484	
2007	64,392.59	0.136248		2.686423	
2008	72,382.84	0.116971		2.838979	
2009	79,701.87	0.096324		3.033187	
2010	87,281.35	0.090844		3.091762	
2011	94,760.76	0.082219		3.191522	
2012	101,821.9	0.07187		3.326046	
2013	107,333.9	0.052719		3.635918	
2014	112,889.1	0.050461		3.679698	
2015	118,303.7	0.046849		3.753965	
2016	123,620.6	0.043962		3.817572	
2017	4282.175		0.160874		2.890937
2018	7786.895	0.597981		1.207343	
2019	10,379.03	0.287345		1.940219	
2020	11,899.95	0.136747		2.682767	
2021	13,163.48	0.100912		2.986651	
2022	14,275.9	0.081126		3.204898	
2023	15,320.8	0.070638		3.343328	
2024	16,354.31	0.06528		3.422217	
2025	17,319.89	0.057364		3.551478	
2026	18,216.46	0.05047		3.679527	

illustrated that VSSGM is a good fitting model, showing an accurate prediction. In Fig. 4, the linear trend indicates the amount of CO₂ that will increase upward gradually until 2026 for China is aligned with the findings of Xie et al. (2015) and Lakshmi and Visalakshmi (2016), in terms of the measurement procedure. Likewise, in the case of the USA, the growth rate of CO₂ emissions gradually decreases until 2026, with an increasing rate of electricity production by renewable resources. China has the highest renewable energy consumption with 106.7 million tons of oil-equivalent, contributing with 22% in the total world renewable energy consumption. China also has the highest CO₂ emissions volume with 466.23 million tons, contributing to the major portion of total world CO₂ emissions (US Energy Information Administration 2017).

The RGR for China, in terms of electricity production by renewable resources, is 0.677, as shown in Table 4. Comparatively, China's RGR is higher than the USA's at 0.225 (Table 12), which implies that, until 2026, China will be producing electricity based on renewable sources even

more than the USA. Additionally, analyses showed that the USA has less relative growth of CO₂ emissions, with 0.161, as presented in Table 13, if compared with China, at 0.264 (see Table 14). Similarly, USA showed 0.329929 relative growth rate of production of electricity by renewable energy sources comparatively less than China, presented in Table 15. The VSSGM is effective when dealing with China's data, and these results are comparable to the studies of Zhang et al. (2015) and Fang et al. (2018), along with Zhang et al. (2012) and McKibbin et al. (2007). The total amount of electricity generation for China is higher at 4213.842 Billion KWh by the end of 2026. Data showed that China is the leading country in the world concerning CO₂ emissions, with 12,858,597 mm tons, and the USA showed lower CO₂ emissions if compared with China, with 896.5663 mm tons. Then, the USA contributes 2% of the total CO₂ emissions in the world (US Energy Information Administration 2017). The assessment of CO₂ emissions helps us to conclude that there is a mutually agreed-upon consensus that these are critical factors of environmental degradation, global warming, and

Table 14 China relative growth rate and doubling time model results of CO₂ emission

Years	Aggregate values	Relative growth rate (RGR)	Mean RGR	Doubling time (D _t)	Mean D _t
2000	3523.152				
2001	7218.02	0.717225	0.250825	1.025513	2.384292
2002	11,178.53	0.437415		1.520021	
2003	15,803.99	0.346267		1.753693	
2004	21,179.47	0.29277		1.921515	
2005	27,285.07	0.253307		2.066299	
2006	34,025.03	0.220757		2.203841	
2007	41,243.96	0.192408		2.341283	
2008	48,740.64	0.167008		2.482858	
2009	56,929.4	0.155299		2.555551	
2010	65,708.59	0.143418		2.635141	
2011	75,540.87	0.139444		2.663238	
2012	152,432	0.702045		1.046905	
2013	162,579.5	0.064449		3.435035	
2014	173,060.2	0.062473		3.466176	
2015	183,846.5	0.060461		3.4989	
2016	194,912.8	0.058451		3.532708	
2017	11,323.15		0.263937		2.207542
2018	22,881.77	0.703491		1.044847	
2019	34,656.28	0.415138		1.572291	
2020	46,628.73	0.296738		1.908053	
2021	58,782.67	0.23163		2.155761	
2022	71,103	0.190282		2.352393	
2023	83,575.89	0.161626		2.51562	
2024	96,188.65	0.140556		2.655294	
2025	108,929.7	0.124391		2.777473	
2026	121,788.3	0.111582		2.886146	

climate vulnerability while these uprising levels of environmental deterioration.

The reduction of CO₂ emissions ensures the development of environmental sustainability by decreasing the utilization of non-renewable sources for electricity production growth, which helps to yield the total energy consumption. Consequently, the efficient production of electricity by renewable sources, such as solar, wind, and hydro, could decrease the amount of CO₂ emissions up to 75%. Moreover, the inefficient structuring and improper maintenance of national energy consumption are causing an increase in CO₂ emissions (Şahin 2019; Chen et al. 2020; Dogan 2016). Hence, our analysis uptakes that China and the USA renewable electricity generation will increase by 2026 based on ODGM, NDGM, and VSSGM prediction results. Conversely, the USA showed a declining trend of CO₂ emissions in the future.

When the ODGM is applied to forecast the total quality of CO₂ and renewable electricity production based on actual and forecast data, different sequences of results are obtained. An important issue will be, in the long term, the emissions

scenarios between China and the USA. When trying to shed some light on this issue, the grey synthetic index was applied to past (actual) and future data (forecasting), $RGR_{synthetic}$ showing relative growth rates, as follows:

$$RGR_{synthetic} = \theta(RGR_{past}) + (1-\theta)(RGR_{future}) \quad (34)$$

where RGR_{past} presents the relative growth rate resulted from actual data and RGR_{future} indicates the relative growth rate resulting from future (forecasting) data.

$$D_{synthetic} = \theta(D_{past}) + (1-\theta)(D_{future}) \quad (35)$$

where D_{past} represents the doubling time obtained from actual data and RGR_{future} reports the growth rate obtained from simulated/forecasted data. Here, θ states that the coefficient of relative weights and value of θ can be taken as 0.5. If the decision-makers, while in the decision-making process, take into account the relative growth rate of simulated data more influential, then $\theta < 1 - \theta$.

Table 15 The US relative growth rate and doubling time model results of electricity production

Years	Aggregate values	Relative growth rate (RGR)	Mean RGR	Doubling time (D_t)	Mean D_t
2000	81.01259				
2001	163.8165	0.704142	0.22536	1.043922	2.314844
2002	256.6054	0.448793		1.494342	
2003	350.3151	0.311294		1.860166	
2004	447.8358	0.245594		2.097224	
2005	548.3178	0.202428		2.290516	
2006	658.3013	0.182808		2.392468	
2007	776.4527	0.165073		2.494515	
2008	915.4547	0.164685		2.496867	
2009	1073.137	0.15892		2.5325	
2010	1255.494	0.156943		2.545019	
2011	1467.32	0.155909		2.551632	
2012	1705.369	0.150344		2.587977	
2013	1980.596	0.149616	0.329929	2.592828	1.910243
2014	2284.503	0.142751		2.639804	
2015	2607.831	0.13237		2.715299	
2016	2982.03	0.134086		2.702424	
2017	431.1444				
2018	921.8903	0.759983		0.967606	
2019	1481.143	0.474143		1.439393	
2020	2119.139	0.358196		1.719823	
2021	2847.643	0.295482		1.912295	
2022	3680.178	0.25647		2.053892	
2023	4632.29	0.23009		2.162432	
2024	5721.844	0.21124		2.247909	
2025	6969.376	0.197235		2.316509	
2026	8398.491	0.186526		2.37233	

According to the grey synthetic relative growth rate model, China showed a higher quantity of CO₂ emissions over the USA, as presented in Table 16. Similarly, the same sequence was obtained for electricity generation through renewable sources by synthetic RGR. Our findings showed that, in the long term, the USA could have a competitive edge over China (see Fig. 11(a)). Within this study, we also applied the double-time model, successfully executed by Paramati et al. (2017) and Ikram et al. (2019). Here we find China needs 2.248 number of years to make its renewable electricity double in quantity, as seen in Fig. 11(b).

The USA requires 2.113 number of years to produce electricity double in quantity by a renewable source, as presented in Fig. 11(b). Similarly, China requires 2.845 number of years to decrease CO₂ emissions by two times. The results of this study provide new insights for CO₂ emissions and energy production within two of the top energy-consuming countries globally. Our evidence is supported by scientific consensus that there are solutions for global warming and climate change. We can take the opportunity for these solutions further through their alignment with the

United Nation's 17 Sustainable Development Goals (SDGs) and suggest that both researchers and decision-makers consider this alignment.

Conclusion

This research in this study aims to measure an accurate estimation for forecasting CO₂ emissions and renewable electricity production in the year 2026 by considering data from two of the most polluting countries in the world. We used advanced mathematical optimized discrete grey, nonhomogeneous discrete, and variable speed and adaptive structure grey modeling to obtain accurate forecasts. We employed performance criteria for measuring models' accuracy. In the long term, we can predict that CO₂ emissions will decrease dramatically in the USA. At the same time, China needs to focus more on its electricity production and renewable energy strategy to reduce CO₂ emissions. The proposed methodology is an innovative application in CO₂ emissions and renewable energy forecasting while reducing the uncertainty in

Table 16 RGR and Dt results from the comparison of CO₂ emissions and electricity production for the USA and China

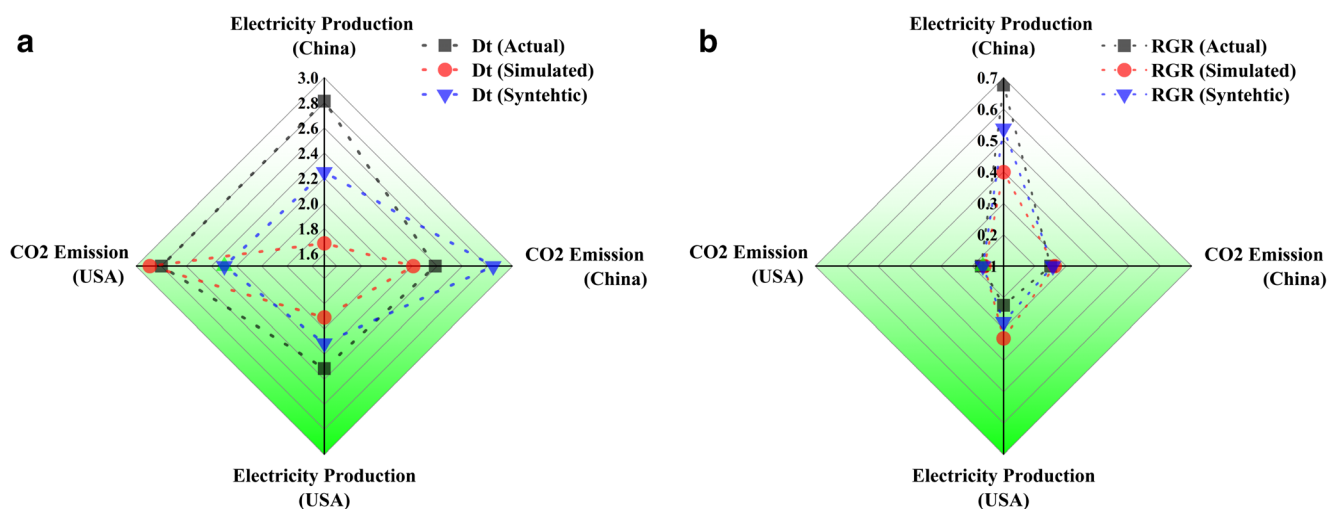
	Country	RGR (Actual data)	D _t (Actual Data)	RGR (Simulated Data)	D _t (Simulated data)	RGR (Synthetic)	D _t (Synthetic)
Electricity Production	USA	0.225	2.315	0.330	1.910	0.278	2.113
	China	0.677	2.813	0.399	1.682	0.538	2.248
CO ₂ Emissions	USA	0.171	2.798	0.161	2.891	0.166	2.296
	China	0.250	2.384	0.263	2.208	0.257	2.845

ambiguity and numerical ranges typically involved in these kinds of studies. The results showed that we can successfully utilize ODGM, NDGM, and VSSGM grey models based on the existing data for forecasting the trends of CO₂ emissions levels. Moreover, the study outcomes confirmed the accuracy of the VSSGM model if compared with two forecastings ODGM and NDGM grey models.

We recommend future studies to advance this area of study by comparing China and the USA since policy-makers need to have accurate data for decision analysis. We suggest that researchers continue using VSSGM grey models in the estimation of similar energy and emission forecasting. We can compare these types of forecasts with parametric versus non-parametric comparisons across countries by using grey forecasting models. As climate change continues, more studies are needed in comparing countries along with the economic development and the prioritization of renewable energy sources at a global scale, to enable better decision-making outcomes.

Policy-makers must address the Sustainable Development Goals (SDGs) recommended by the United Nations when revising the national policies towards sustainability. For

example, the European Union is targeting the Green Deal aiming a balance between growth strategy and resource-efficiency, which encompasses no net emissions of greenhouse gas emissions by the year 2050, and decoupling economic growth from natural resources use (European Commission 2020; European Parliament 2019). The USA can likewise improve its position for the future if it can adopt its own Green New Deal. The achievement of such goals also requires the promotion of renewable energies, e.g., renewable electricity production, among European Union Member States and states within the USA. Countries such as China and the USA need to consider the formation of Sustainable Energy Action Plans, a set of policies devoted to the increase of energy efficiency use and the consideration of renewable energy sources at local levels, as also recommended by the European Commission (2020). Thus, we recommended policy-makers review and develop such initiatives in order to implement policies moving countries toward more sustainable development of energy infrastructure. This development can and should be promoted at the cities-level (Alhorr et al. 2014). This level of implementation can be fast, with the benefits felt earlier by citizens.


Fig. 11 (a) Relative growth rate (RGR) of China Vs. USA. (b) Doubling time (Dt) of China Vs. USA

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Authors' contribution Conceptualization, Muhammad Ikram; formal analysis, Muhammad Ikram; investigation, Muhammad Ikram, Robert Sroufe, Qingyu Zhang, Marcos Ferraso; methodology, Muhammad Ikram; project administration and supervision, Qingyu Zhang; validation, Muhammad Ikram; writing and original draft, Muhammad Ikram; writing, review, and editing, Muhammad Ikram, Robert Sroufe, Qingyu Zhang, and Marcos Ferraso

Data Availability All data generated or analysed during this study are included in this published article.

Compliance with ethical standard

Conflicts of interests The authors declare that they have no conflict of interest.

Ethics approval Not applicable

Consent to participate Not applicable

Consent to publish Not applicable

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